

Security+ 4.1 · Wireless Security Settings

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 4.1** —

Wireless Security Settings (10:54) · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: AP configuration: WPA3/SAE (modern), WPA2/TKIP and WEP are broken and prohibited; centralized RADIUS; WPS (Wi-Fi Protected Setup) is insecure — DISABLE it.

High-Yield Core Concepts Tested on Exam

- WPA3/SAE = modern; WEP/WPA never use
- WPS (the push-button) is INSECURE — always disable
- RADIUS provides centralized authentication
- WPA2/TKIP is deprecated; WPA3/SAE is preferred

Exam Tips & Classic Scenarios

- 'deprecate WEP/WPA' = use WPA2/3
- 'disable WPS' = remove insecure push-button setup
- 'WPA3-SAE' = modern wireless security

Typical Exam Scenario: 'Disable the WPS button that users can press' = WPS is insecure.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 4.1 — Wireless Security Settings)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **RADIUS:** Remote Authentication Dial-In User Service: Client/server AAA protocol (UDP ports 1812/1813) standard in 802.1X and VPN environments; encrypts only the password field and combines authentication and authorization.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. Service that centralizes network access authentication (login to switch/VPN via central server):

- A) RADIUS**
- B) DNS**
- C) SMTP**
- D) NTP**

2. Corporate scenario (Wireless Security Settings): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.1), an analyst evaluates the following technical requirement: 'WPA3/SAE = modern; WEP/WPA never.' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: WPA3/SAE = modern; WEP/WPA never...
- D)** Allow shared access among administrators using generic credentials

3. Corporate scenario (Wireless Security Settings): During a security review and compliance audit aligned with the CompTIA Security+ (SY0-701, Section 4.1) exam, an analyst evaluates the following technical requirement: 'WPS (the little button) is INSECURE — always disable it'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: WPS (the little button) is INSECURE — always disable it...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

4. Corporate scenario (Wireless Security Settings): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.1), an analyst evaluates the following technical requirement: 'deprecate WEP/WPA'=use WPA2/3'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'deprecate WEP/WPA'=use WPA2/3...

5. Corporate scenario (Wireless Security Settings): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.1), an analyst evaluates the following technical requirement: 'route disable the WPS button that the user can press' = WPS insecure. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'route disable the WPS button that the user can press' = WPS insecure...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

6. Corporate scenario (Wireless Security Settings): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.1), an analyst evaluates the following technical requirement: 'WPA3/SAE = modern; WEP/WPA never'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: WPA3/SAE = modern; WEP/WPA never...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

7. Corporate scenario (Wireless Security Settings): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.1), an analyst evaluates the following technical requirement: 'WPS (the little button) is INSECURE — always disable it'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: WPS (the little button) is INSECURE — always disable it.
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

8. Corporate scenario (Wireless Security Settings): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.1), an analyst evaluates the following technical requirement: 'deprecate WEP/WPA'=use WPA2/3'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'deprecate WEP/WPA'=use WPA2/3

9. Corporate scenario (Wireless Security Settings): During a security review and compliance audit for the CompTIA Security+ exam (SY0-701, Section 4.1), an analyst evaluates the following technical requirement: 'route disable the WPS button that the user can press'=WPS insecure.'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal adherence terms
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'route disable the WPS button that the user can press'=WPS insecure....

10. Corporate scenario (Wireless Security Settings): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.1), an analyst evaluates the following technical requirement: 'WPA3/SAE = modern; WEP/WPA never'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: WPA3/SAE = modern; WEP/WPA never...

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) A — RADIUS.

✓ **Why it is correct:** RADIUS (Remote Authentication Dial-In User Service) is an authentication, authorization, and accounting (AAA) protocol that centralizes network access control. Access devices such as switches, routers, VPNs, and wireless access points act as RADIUS clients and forward user credentials to a central RADIUS server. This server verifies credentials against a directory (e.g., Active Directory, LDAP) and returns an accept or reject decision, along with authorization attributes (e.g., VLAN, session timeout). This allows consistent access policy management across the infrastructure without configuring local accounts on each device.

✗ **Why the others are wrong:**

- *DNS:* DNS (Domain Name System) is a name resolution service that translates human-readable domain names (e.g., www.example.com) into IP addresses. It does not authenticate users, manage access sessions, or centralize credentials. Although DNSSEC exists to authenticate DNS responses, that is security of the protocol itself, not a network access authentication mechanism for users or devices. Therefore, DNS does not meet the requirement of centralizing login on switches or VPNs.
- *SMTP:* SMTP (Simple Mail Transfer Protocol) is the standard protocol for sending and transferring email messages between servers. It handles asynchronous email communication, not network access authentication. Although SMTP AUTH exists for sending emails, it applies to the email service itself, not to access control for switches, VPNs, or other network equipment. Therefore, it is completely inapplicable to the described scenario.
- *NTP:* NTP (Network Time Protocol) is used to synchronize device clocks on a network with a trusted time source. Precise synchronization is important for forensic logs, certificate validity, and time-based authentication protocols (such as Kerberos), but NTP itself does not authenticate users or centralize network access. It does not receive credentials, query directories, or make authorization decisions. Therefore, it does not replace a RADIUS server.

2) C — Implement the corresponding solution: WPA3/SAE = modern; WEP/WPA never....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.1 (Wireless Security Settings): WPA3/SAE = modern; WEP/WPA never. WPA3 with Simultaneous Authentication of Equals (SAE) provides forward secrecy and resistance to offline dictionary attacks, whereas WEP and WPA (TKIP) are deprecated due to well-known cryptographic weaknesses (e.g., RC4 keystream reuse, weak IVs).

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

3) B — Implement the corresponding solution: WPS (the little button) is INSECURE — always disable it....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.1 (Wireless Security Settings): WPS (the little button) is INSECURE — always disable it. Wi-

Fi Protected Setup (WPS) has known vulnerabilities (e.g., brute-force PIN attacks) that can allow attackers to recover the network's pre-shared key, so it should be disabled.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

4) D — Implement the corresponding solution: 'deprecate WEP/WPA'=use WPA2/3....

✓ Why it is correct: According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.1 (Wireless Security Settings): 'deprecate WEP/WPA'=use WPA2/3. WEP and WPA are outdated and vulnerable to attacks (e.g., IV collisions, TKIP weaknesses). WPA2 with AES-CCMP and WPA3 with SAE provide stronger encryption and authentication, and should be used instead.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

5) A — Implement the corresponding solution: 'route disable the WPS button that the user can press' = WPS insecure....

✓ Why it is correct: According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.1 (Wireless Security Settings): 'route disable the WPS button that the user can press' = WPS insecure. WPS (Wi-Fi Protected Setup) is considered insecure because its PIN-based authentication method is vulnerable to brute-force attacks, allowing attackers to recover the WPS PIN and subsequently the network's WPA/WPA2 pre-shared key. Disabling the WPS button prevents users from enabling this insecure feature, thereby reducing the attack surface. This is a preventive technical control that enforces secure configuration.

✗ Why the other options are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

6) B — Implement the corresponding solution: WPA3/SAE = modern; WEP/WPA never....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.1 (Wireless Security Settings): WPA3/SAE = modern; WEP/WPA never.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

7) B — Implement the corresponding solution: WPS (the little button) is INSECURE — always disable it..

✓ **Why it is correct:** Per the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.1 (Wireless Security Settings): WPS (the little button) is INSECURE — always disable it. WPS (Wi-Fi Protected Setup) was designed to simplify connecting devices to a wireless network, but its PIN-based method is vulnerable to brute-force attacks (the WPS PIN can be cracked in hours), and even the push-button method can be abused. Security+ guidance is to disable WPS entirely and instead use WPA2/WPA3 with strong passphrases or enterprise authentication (802.1X).

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaches compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

8) D — Implement the corresponding solution: 'deprecate WEP/WPA'=use WPA2/3.

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.1 (Wireless Security Settings): 'deprecate WEP/WPA'=use WPA2/3. WEP and WPA are outdated and vulnerable; WPA2/WPA3 provide stronger encryption and authentication.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

9) D — Implement the corresponding solution: 'route disable the WPS button that the user can press'=WPS insecure.....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.1 (Wireless Security Settings): 'route disable the WPS button that the user can press'=WPS insecure..

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal adherence terms:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

10) D — Implement the corresponding solution: WPA3/SAE = modern; WEP/WPA never....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.1 (Wireless Security Settings): WPA3/SAE = modern; WEP/WPA never.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.